

HiveForce Labs

THREAT ADVISORY



ATTACK REPORT

China-Linked Operators Breach Japanese Shipping Networks

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Admiralty Code

A1

TA Number

TA2025374

Summary

Attack Discovered: April 2025

Targeted Country: Japan

Targeted Industries: Shipping, Transportation

Targeted Platform: Windows

Malware: MetaRAT, Talisman PlugX

Attack: A China-based threat group infiltrated Japanese shipping and transportation networks by exploiting two critical Ivanti Connect Secure vulnerabilities, CVE-2024-21893 and CVE-2024-21887, which allowed them to slip past defenses and establish a silent foothold. Using these flaws, the attackers moved from initial access to deeper reconnaissance, stole privileged credentials, and deployed advanced PlugX variants like MetaRAT and Talisman across internal servers. Their operation combined stealthy DLL side-loading, encrypted payloads, and persistent backdoor access, reflecting a well-practiced, multi-stage intrusion designed not for immediate data theft but for long-term espionage and future re-entry.

Attack Regions



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CVE	NAME	AFFECTED PRODUCT	ZERO-DAY	CISA KEV	PATCH
CVE-2024-21893	Ivanti Connect Secure, Policy Secure, and Neurons Server-Side Request Forgery (SSRF) Vulnerability	Ivanti Connect Secure, Ivanti Policy Secure and Ivanti Neurons for ZTA			
CVE-2024-21887	Ivanti Connect Secure and Policy Secure Command Injection Vulnerability	Ivanti Connect Secure and Policy Secure			

Attack Details

#1

A China-based threat group launched a coordinated attack campaign in April 2025, targeting Japanese shipping and transportation companies. The attackers exploited vulnerabilities in Ivanti Connect Secure (ICS), specifically CVE-2024-21893 and CVE-2024-21887, to break into corporate networks and deploy advanced variants of the PlugX malware family, including MetaRAT and Talisman PlugX. These operations highlight the continued evolution of PlugX-based tooling and show how Chinese espionage groups are still relying on refined versions of long-standing malware to infiltrate high-value sectors.

#2

Once inside, the attackers used the Ivanti flaws to install malware that helped them map the network, steal credentials, and move laterally through internal systems. During forensic analysis, investigators also uncovered several lightweight malware files, LITTLELAMB, WOOLTEA, PITSOCK, and PITFUEL, flagged by Ivanti’s Integrity Checker Tool (ICT). These implants are commonly seen in campaigns abusing the same Ivanti issues and are typically used for early-stage reconnaissance, persistence, and network discovery. Their presence, along with a spike in critical “ERR31093” error logs linked to malicious SAML payloads, provided clear markers of the initial intrusion and reinforced that the attackers used a familiar toolkit to establish a foothold before deploying more advanced RATs.

#3

PlugX itself is a long-running remote access Trojan that has spawned numerous variants since 2008 and remains widely used by China-aligned groups such as Calypso, Mustang Panda, and APT10. In this incident, two variants emerged as key components: MetaRAT, a modernised iteration with enhanced obfuscation, modularity, and encrypted C2 communications, and Talisman PlugX, a side-loading variant capable of executing multiple plugins for tasks like keylogging, file manipulation, and command execution. Both variants follow the traditional PlugX model but exhibit sharper code changes, refined encryption routines, and new command formats that help them evade detection.

#4

MetaRAT stood out for its sophisticated execution chain. It relies on DLL side-loading, custom shellcode, layered decryption, and reflective loading to unpack itself directly into memory. The malware supports multiple communication protocols, including TCP, UDP, HTTP, HTTPS, and ICMP, and uses AES-based encryption and compressed payloads to securely exchange data with its command-and-control servers. Talisman follows a similar flow, loading an encrypted payload ("ws32.sob"), decrypting and decompressing embedded components, and injecting itself into legitimate processes to blend into normal system activity. Both variants also share certain code characteristics, suggesting they may originate from overlapping developer ecosystems.

#5

Taken together, the evidence points toward involvement by China-linked groups such as Space Pirates or Calypso, with some indications of shared tooling or infrastructure used previously by RedFoxtro. While no data theft was observed, the attackers clearly focused on credential harvesting and establishing long-term access for future operations. This campaign underscores the importance of strong patch management, continuous monitoring, and multi-layered security controls, especially for organizations running exposed VPN appliances or remote access infrastructure.

Recommendations



Patch Ivanti Devices Without Delay: Apply the latest security updates for Ivanti Connect Secure and any related appliances as soon as they are released. These vulnerabilities were the attackers' entry point, so closing them immediately is essential to prevent similar breaches.



Strengthen Monitoring Around ICS and VPN Gateways: Continuously monitor for unusual login attempts, spikes in error logs such as ERR31093, or unexpected file creation on Ivanti devices. Early detection in these areas could stop attackers before they move deeper into the network.



Review Internal Network Segmentation: Separate critical servers so that even if an attacker steals credentials, they cannot move across the entire network. Create clear boundaries between ICS, business systems, and sensitive data stores.



Build Incident Response Playbooks For Malware Like Plugx Variants: Prepare procedures for identifying, isolating, and removing RATs such as PlugX, MetaRAT, and Talisman. Train teams on recognizing common indicators, side-loaded DLLs, encrypted payloads, and modified PE headers.



Enhance Endpoint Protection: Deploy next-generation antivirus (NGAV) and endpoint detection & response (EDR) solutions to identify and block malware. Leverage behavioral analysis and machine learning-based detection to spot suspicious activity.

Potential MITRE ATT&CK TTPs

<u>TA0042</u> Resource Development	<u>TA0001</u> Initial Access	<u>TA0002</u> Execution	<u>TA0003</u> Persistence
<u>TA0005</u> Defense Evasion	<u>TA0007</u> Discovery	<u>TA0009</u> Collection	<u>TA0011</u> Command and Control
<u>T1190</u> Exploit Public-Facing Application	<u>T1588</u> Obtain Capabilities	<u>T1588.006</u> Vulnerabilities	<u>T1078</u> Valid Accounts
<u>T1574</u> Hijack Execution Flow	<u>T1574.001</u> DLL	<u>T1027</u> Obfuscated Files or Information	<u>T1140</u> Deobfuscate/Decode Files or Information
<u>T1055</u> Process Injection	<u>T1106</u> Native API	<u>T1071</u> Application Layer Protocol	<u>T1071.001</u> Web Protocols
<u>T1095</u> Non-Application Layer Protocol	<u>T1056</u> Input Capture	<u>T1056.001</u> Keylogging	<u>T1082</u> System Information Discovery
<u>T1543</u> Create or Modify System Process	<u>T1543.003</u> Windows Service	<u>T1547</u> Boot or Logon Autostart Execution	<u>T1547.001</u> Registry Run Keys / Startup Folder
<u>T1070</u> Indicator Removal	<u>T1070.006</u> Timestomp	<u>T1059</u> Command and Scripting Interpreter	

Indicators of Compromise (IOCs)

TYPE	VALUE
SHA256	8f50f434fa776cd11869d1e43858ac6d480d176b7c0dddc09b71d40b20a4aa46, c6124a3ca27299ef1b4d053782b11fe9fe4e215cac3cbd47d09a06e2ca6dde0c, 6ad67d7f76986359865667bdd51ba267f6bd7e560270512074448dd7b088bcb7,

TYPE	VALUE
SHA256	00dbc8a4b3121af5a19504a9d969e36e709556420a6117eb3533f1d2a8100fd9, aec2d0cbd2f195bf35e55019a29f0d6109451eb85dc7941b73e3b562b065a11c, fd87149d6b8fdcad5d84ba4a3ca52e1cef8f0c54cafca6dbbb5d156f313d79dd, fd6b1ca0f26e54fa9c97ea15c834e58ffb71798df38071ad00b14f19d6a4126c, c91595edd1c9a0a2c1168e3bfa532e4a7dbb6b1380afd80ba445b728622798a4, c90460e820a8c5874d5412032b7db719cb8ea34ae8e48e4ab934a4096a09612b, a92ed5f831c99bb84208ef7d7c733e0183a79de40f9d3b3be54744951f0a1391, 0ec83d1deb6065cac8ba8f849cdf5672da7313ec2e860a7d71bb7e397e661394, 7b028a9bd2bc0c306ab6561cf702406f5925fc073f9d0d2d9408ceccd6907743, aba6f7611291433983ba9c65654b04745a050530329d3ad329cc859c1ce12c44, d3ec33ae5c8ce2ac5eb0c96c6d6dc1d5ca610bacaa9de85d1e4bfe1d60923970, ae0f29727920882781d31d982c0b5ad71d83f1dd9cebd41e1e5e444c1e6d7854, 78c3eb67fdc59fd09cba6388d6e31c428ed3c227f04b9cd739e8c36a8f1a182e, 367ad2eaa851ae17a4b75d92ec712d889fa85c0f2a51b9d5c5e08ae84fa7514d, 88bf17cb9f126c9c5a78a2a29138ee1cdaab3ec141115f45a1f345bb6679d57d, Df5b319cc7e1f4a6389a02b7942b5db68ca55eaeb7ad8d7b78f0ab6671496d66
Domains	doodle01[.]space, piao[.]mil[.]onmypc[.]net, newsinfom[.]org, mailserver[.]kozow[.]com, turky[.]info, nord[.]ocry[.]com
IPv4	117[.]254[.]105[.]200, 45[.]114[.]192[.]137, 103[.]9[.]14[.]218, 23[.]254[.]225[.]184, 103[.]136[.]45[.]108, 103[.]172[.]10[.]165, 117[.]239[.]199[.]202, 220[.]130[.]204[.]242, 112[.]213[.]125[.]75

Patch Links

CVE-2024-21893:

https://forums.ivanti.com/s/article/CVE-2024-21888-Privilege-Escalation-for-Ivanti-Connect-Secure-and-Ivanti-Policy-Secure?language=en_US

CVE-2024-21887:

https://forums.ivanti.com/s/article/CVE-2023-46805-Authentication-Bypass-CVE-2024-21887-Command-Injection-for-Ivanti-Connect-Secure-and-Ivanti-Policy-Secure-Gateways?language=en_US

References

https://www.lac.co.jp/lacwatch/report/20251208_004569.html

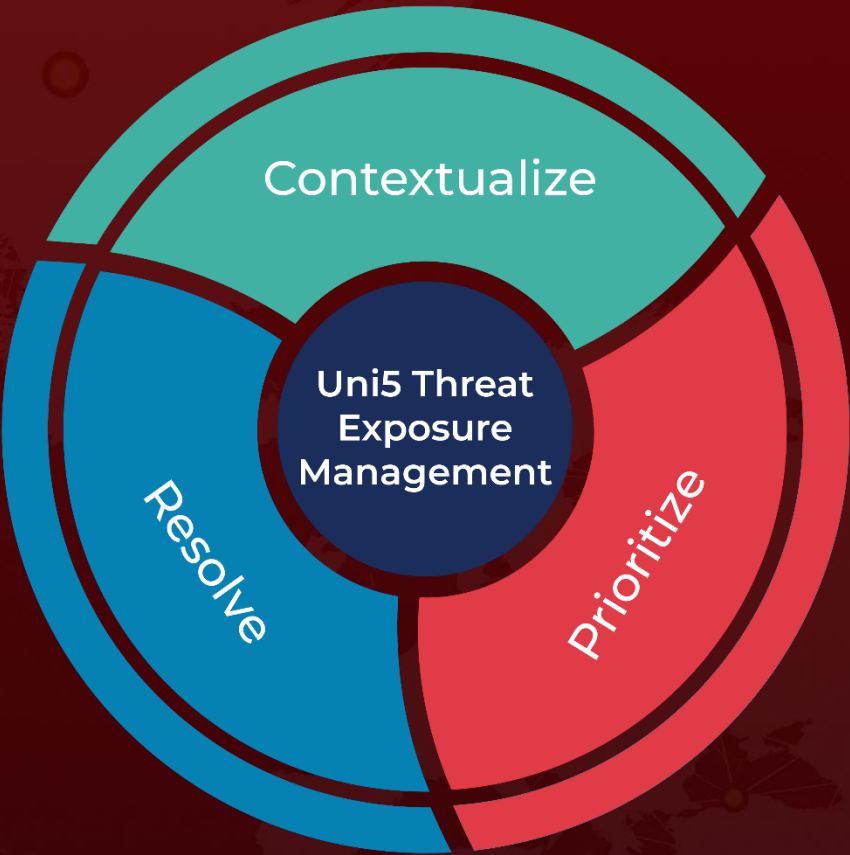
<https://www.hivepro.com/threat-advisory/ivanti-addresses-zero-day-vulnerability-exploited-in-attacks/>

<https://www.hivepro.com/threat-advisory/ivanti-gateways-under-attack-by-cybercriminals-patch-now/>

What Next?

At Hive Pro, it is our mission to detect the most likely threats to your organization and to help you prevent them from happening.

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